SQL Server MERGE

The `MERGE` **statement handles** `UPDATE`**,** `INSERT`**, and** `DELETE` **all in one single query.**

🔹 What each part means in `MERGE`

- `target_table`

  👉 This is your **main / existing table in the database**

  (the one you want to update, insert into, or delete from)

- `source_table`

  👉 This is the **new data / incoming data**

  (can be a table, staging table, subquery, or even a CTE)

🔹 Simple way to think about it

Term	Meaning
`target_table`	Existing data (already stored)
`source_table`	New data (coming in for comparison)

🔹 What happens in each condition

- **WHEN MATCHED**

  👉 If a row exists in both target & source → **UPDATE**

- **WHEN NOT MATCHED**

  👉 If row exists in source but NOT in target → **INSERT**

- **WHEN NOT MATCHED BY SOURCE**

  👉 If row exists in target but NOT in source → **DELETE**

SYNTAX:

```sql
MERGE target_table
USING source_table
ON merge_condition
WHEN MATCHED
    THEN update_statement
WHEN NOT MATCHED
    THEN insert_statement
WHEN NOT MATCHED BY SOURCE
    THEN DELETE;
```

```
69  SELECT * FROM CRICKETER_DETAIL
70
```

BEFORE MERGE

121 % 6 0

Results | Messages

	ID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS
1	1	Rohit	Sharma	1987-04-30	Batsman	121
2	2	Virat	Kohli	1988-11-05	Batsman	122
3	3	Sanju	Samson	1994-11-11	Wicketkeeper-Batsman	111
4	4	Jasprit	Bumrah	1993-12-06	Bowler	16
5	5	Shivam	Dube	1993-06-26	All-rounder	63

```sql
DECLARE @SOURCE_JSON VARCHAR(MAX) ='
[
  {
    "ID":1,
    "FIRST_NAME": "ROHIT",
    "LAST_NAME": "SHARMA",
    "DOB": "1987-04-30",
    "ROLE": "Batter",
    "HIGHEST_T20I_RUNS": 121
  },
  {
    "ID":2,
    "FIRST_NAME": "VIRAT",
    "LAST_NAME": "KOHLI",
    "DOB": "1988-11-05",
    "ROLE": "Batter",
    "HIGHEST_T20I_RUNS": 122
  },
    {
      "ID":10,
      "FIRST_NAME": "Abhishek",
      "LAST_NAME": "Sharma",
      "DOB": "2000-09-04",
      "ROLE": "Batting All-rounder",
      "HIGHEST_T20I_RUNS": 100
    },
    {
      "ID":11,
      "FIRST_NAME": "Yashasvi",
      "LAST_NAME": "Jaiswal",
      "DOB": "2001-12-28",
      "ROLE": "Opening Batter",
      "HIGHEST_T20I_RUNS": 100
    },
    {
      "ID":12,
      "FIRST_NAME": "Rinku",
      "LAST_NAME": "Singh",
      "DOB": "1997-10-12",
      "ROLE": "Middle-order Batter",
      "HIGHEST_T20I_RUNS": 69
    },
    {
      "ID":13,
      "FIRST_NAME": "Axar",
      "LAST_NAME": "Patel",
      "DOB": "1994-01-20",
      "ROLE": "All-rounder",
      "HIGHEST_T20I_RUNS": 65
    }
  ]'
```

```sql
DROP TABLE IF EXISTS #SOURCE;
```

```sql
CREATE TABLE #SOURCE(
    ID INT,
    FIRST_NAME VARCHAR(20),
    LAST_NAME VARCHAR(20),
    DOB DATE,
    ROLE VARCHAR(50),
    HIGHEST_T20I_RUNS INT
);
```

```sql
INSERT INTO #SOURCE(ID,FIRST_NAME,LAST_NAME,DOB,ROLE,HIGHEST_T20I_RUNS)
SELECT
    TRY_CAST(ID AS INT),
    FIRST_NAME,
    LAST_NAME,
    TRY_CAST(DOB AS DATE),
    ROLE,
    TRY_CAST(HIGHEST_T20I_RUNS AS INT)
FROM OPENJSON(@SOURCE_JSON)
WITH(
    ID VARCHAR(20),
    FIRST_NAME VARCHAR(20),
    LAST_NAME VARCHAR(20),
    DOB VARCHAR(20),
    ROLE VARCHAR(50),
    HIGHEST_T20I_RUNS VARCHAR(20)
);
```

```sql
SELECT * FROM #SOURCE;
```

	ID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS
1	1	ROHIT	SHARMA	1987-04-30	Batter	121
2	2	VIRAT	KOHLI	1988-11-05	Batter	122
3	10	Abhishek	Sharma	2000-09-04	Batting All-rounder	100
4	11	Yashasvi	Jaiswal	2001-12-28	Opening Batter	100
5	12	Rinku	Singh	1997-10-12	Middle-order Batter	69
6	13	Axar	Patel	1994-01-20	All-rounder	65

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TARGET TABLE

	ID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS
1	1	Rohit	Sharma	1987-04-30	Batsman	121
2	2	Virat	Kohli	1988-11-05	Batsman	122
3	3	Sanju	Samson	1994-11-11	Wicketkeeper-Batsman	111
4	4	Jasprit	Bumrah	1993-12-06	Bowler	16
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SOURCE TABLE

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```sql
MERGE CRICKETER_DETAIL AS T
USING #SOURCE AS S
ON T.ID = S.ID
WHEN MATCHED
    THEN
        UPDATE
        SET
            T.FIRST_NAME = S.FIRST_NAME,
            T.LAST_NAME = S.LAST_NAME,
            T.DOB = S.DOB,
            T.ROLE = S.ROLE,
            T.HIGHEST_T20I_RUNS = S.HIGHEST_T20I_RUNS
WHEN NOT MATCHED
    THEN
        INSERT (FIRST_NAME,LAST_NAME,DOB,ROLE,HIGHEST_T20I_RUNS)
        VALUES (S.FIRST_NAME,S.LAST_NAME,S.DOB,S.ROLE,S.HIGHEST_T20I_RUNS)
WHEN NOT MATCHED BY SOURCE
    THEN
        DELETE;
```

```
SELECT * FROM CRICKETER_DETAIL;
```

AFTER MERGE

91 % No issues found

Results | Messages

	ID	FIRST_NAME	LAST_NAME	DOB	ROLE	HIGHEST_T20I_RUNS
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2	2	VIRAT	KOHLI	1988-11-05	Batter	122
3	6	Abhishek	Sharma	2000-09-04	Batting All-rounder	100
4	7	Yashasvi	Jaiswal	2001-12-28	Opening Batter	100
5	8	Rinku	Singh	1997-10-12	Middle-order Batter	69
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BEFORE MERGE ANS

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